

K-FIELD FLYWHEEL FORCE ANALYSIS

Exact Continuous-Rim Integration and Full Orthogonal-Axis Sweep About the
Earth-CMB Vector

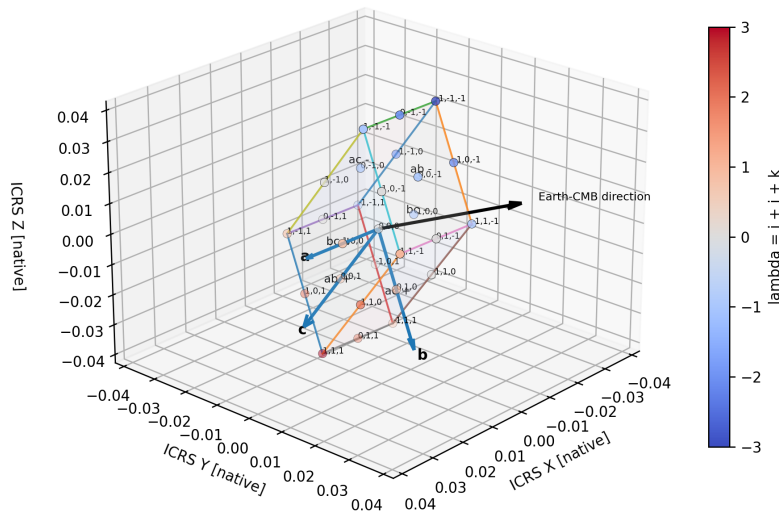
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ICRS Cell3 base cell: 27 nodes and six boundary face planes



Title figure. Authoritative ICRS Cell3 base cell in native coordinates, including all 27 centered nodes, all six boundary face planes, the primitive basis vectors, and the Earth-CMB direction. Every coordinate is generated directly from the attached machine-readable source.

Archival computational edition - Unix epoch 1784808116

Abstract

This reference develops the exact net-force response of a one-metre-diameter, one-kilogram, uniform zero-thickness flywheel rim located at Earth center in the attached ICRS K-Field frame. The flywheel translates with the Earth-CMB velocity vector, and its directed angular-velocity axis is constrained to remain orthogonal to that vector while the axis is swept through a complete 360-degree rotation about the Earth-CMB direction. The elemental K-Field force is integrated over the continuous mass distribution rather than approximated by isolated point masses. The R omega-squared rotational self-term cancels exactly under antipodal rim integration. The surviving translation-coupled resultant is $F(\alpha)=(\kappa_F M \omega/2) \mathbf{n}(\alpha) \text{ cross } [G_{\text{phase}} \mathbf{V}_E]$. The vector therefore traces a centered ellipse in the ICRS plane perpendicular to $G_{\text{phase}} \mathbf{V}_E$. For $M=1 \text{ kg}$ and the normalized reference $\omega=1 \text{ rad/s}$, the semi-major force is 465412.608635 N per unit κ_F , the semi-minor force is 396230.434489 N per unit κ_F , the constant magnitude perpendicular to the Earth-CMB direction is 396230.434489 N per unit κ_F , and the Earth-CMB-parallel component has amplitude 244152.286620 N per unit κ_F . The full-cycle mean force is exactly zero, the local and net torque about the flywheel center are exactly zero, and reversing spin reverses the instantaneous force vector without changing the geometric envelope. The numerical companion file contains the source geometry, all 27 Cell3 nodes, all vertices and face planes, the exact formulas, the complete 0.1-degree sweep ledger, extrema, scaling laws, and numerical reconstruction checks.

Field	Authoritative or derived value
Coordinate frame	ICRS; origin at Earth center
Flywheel	Uniform zero-thickness rim; diameter 1.0 m; radius 0.5 m; total mass 1.0 kg
Earth-CMB velocity	[-359003.199066, 76679.886034, -44701.060101] m/s; speed 369812.502000 m/s
Axis constraint	$\mathbf{n}(\alpha) \cdot \mathbf{c}_{\text{hat}} = 0$; $\mathbf{n}(\alpha)=\cos(\alpha)\mathbf{e}_0+\sin(\alpha)\mathbf{A}_0$
Exact resultant	$\mathbf{F}=(\kappa_F M \omega/2) \mathbf{n} \text{ cross } (G_{\text{phase}} \mathbf{V}_E)$
Envelope	Centered ellipse; semi-major 465412.608635; semi-minor 396230.434489 N/($\kappa_F \text{ rad/s}$)
Torque	Zero at every rim element and therefore zero after integration
Absolute normalization	All force values scale linearly with the source-defined constitutive gain κ_F
Machine-readable companion	K-Field_flywheel_force_analysis_1784808116.json

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1. Problem Definition and Scope

The objective is to determine the complete net-force vector envelope of a simple continuous flywheel moving with the Earth-CMB translation state in the attached K-Field geometry. The calculation uses a fixed ICRS coordinate frame and places the flywheel center at Earth center. The physical body is a circular rim with diameter 1.0 m, radius 0.5 m, total mass 1.0 kg, zero axial thickness, and uniform linear mass density. The flywheel spin axis is always orthogonal to the Earth-CMB direction. Its orientation is swept through a complete directed 360-degree cycle around that direction.

The word directed is essential. A mechanical shaft line is unchanged when its arrow is reversed, but an angular-velocity vector includes spin sense. The parameter $n(\alpha)$ is therefore the directed angular-velocity axis. Replacing n by $-n$ is equivalent to reversing the spin direction at the same shaft orientation. The 360-degree sweep records the complete signed force locus; the underlying shaft geometry repeats after 180 degrees while the force changes sign.

Quantity	Symbol	Value
Total rim mass	M	1.000000 kg
Diameter	D	1.000000 m
Radius	R	0.500000 m
Circumference	$2 \pi R$	3.141592653590 m
Linear mass density	$M/(2 \pi R)$	0.318309886184 kg/m
Thin-rim moment of inertia	$I = M R^2$	0.250000000000 kg m ²
Reference spin rate	ω	1.000000 rad/s = 9.549296586 rpm
Reference rim speed	$R \omega$	0.500000000000 m/s
Reference rotational energy	$0.5 I \omega^2$	0.125000000000 J
Reference angular momentum	$I \omega$	0.250000000000 kg m ² /s

No discretization of the rim is required for the final result. The continuous integral closes analytically. Discrete quadrature is used only as an independent numerical reconstruction of that closed form.

2. ICRS Coordinate Lock and Earth-CMB Translation State

The source fixes a right-handed orthonormal triad composed of e_0 , A_0 , and $c_{\hat{}}$. The Earth-CMB direction is $c_{\hat{}}$. The vectors e_0 and A_0 span the plane perpendicular to $c_{\hat{}}$ and therefore form the natural sweep basis for the flywheel axis. Their orientation obeys $e_0 \text{ cross } A_0 = c_{\hat{}}$.

Vector	ICRS components	RA [deg]	Dec [deg]
$c_{\hat{}}$	[-0.970770855836, 0.207348009111, -0.120874929482]	167.943299880	-6.942599922
e_0	[-0.200864884110, -0.426231479950, 0.882031758970]	244.767479396	61.888432983
A_0	[0.131366829113, 0.880530254232, 0.455422032396]	81.514594988	27.092091822
$n(\alpha) = \cos(\alpha) e_0 + \sin(\alpha) A_0$ $n(\alpha) \cdot c_{\hat{}} = 0$ $e_0 \text{ cross } A_0 = c_{\hat{}}$			

The Earth-CMB translation vector is parallel to $c_{\hat{}}$ and has the ICRS components shown below. Its magnitude is 369.812502 km/s. The K-Field phase metric transforms that velocity into the anisotropic kernel vector $g = G_{\text{phase}} V_E$. The force envelope is controlled by g rather than by V_E alone.

Quantity	ICRS components or value
V_E [m/s]	[-359003.199065813, 76679.886034002, -44701.060100688]

Quantity	ICRS components or value
$ V_E $ [m/s]	369812.502000406443
G_phase V_E [m/s]	[-880221.981635091, -143911.436561599, 266335.026948619]
$ G_{\text{phase}} V_E $ [m/s]	930825.217270434834
Angle between c_{hat} and G_phase V_E	31.640861271139 deg

ICRS directions: Earth-CMB sweep frame and Cell3 face normals

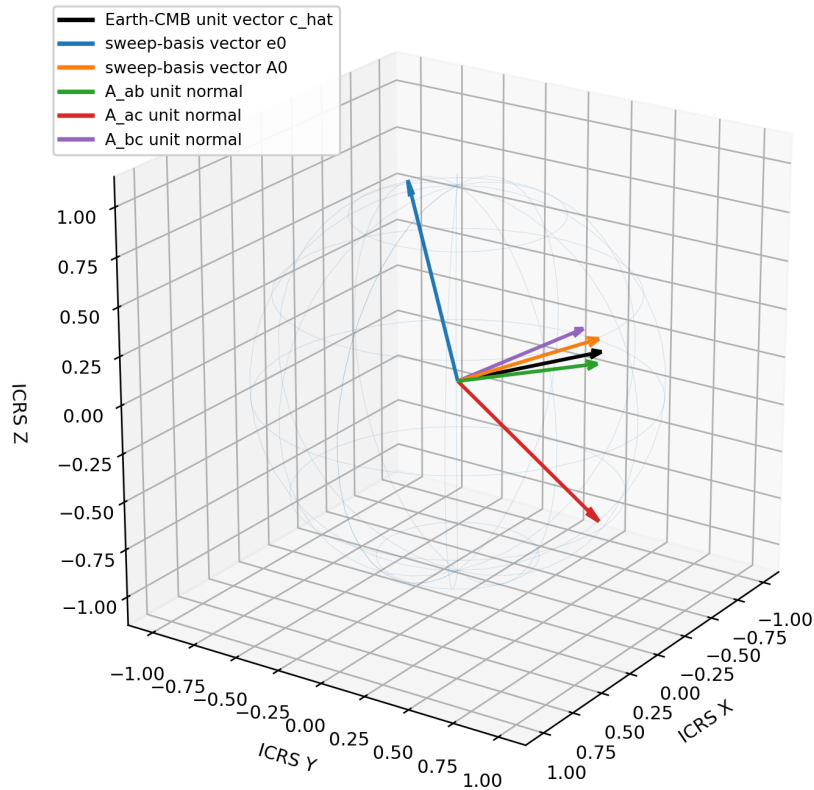


Figure 1. ICRS unit-direction view of the orthonormal Earth-CMB sweep frame and the three authoritative Cell3 face normals. Opposite boundary planes use the same unit normal with opposite offsets.

3. Authoritative Cell3 Base Geometry

The attached source embeds the Cell3 primitive basis directly in ICRS coordinates. The basis columns a, b, and c define the centered parallelepiped used as the geometric reference cell. The cell vertices are all combinations of plus or minus one-half of the three basis vectors. The 27 centered node positions are the combinations of coefficients -1/2, 0, and +1/2 on each basis vector.

Basis vector	ICRS native components	Length [native]
a	[0.020381438516946, -0.003416414496411, -0.002332965916729]	0.020797058782433
b	[0.002198745968323, 0.012902170258693, -0.034066501536056]	0.036494205130727
c	[-0.000630379485347, -0.024257722017128, -0.042924354571094]	0.049308565899859
Cell quantity	Value	
Primitive signed determinant $\det[a \ b \ c]$	-2.842066937809649e-05 native ³	

Cell quantity	Value
Primitive volume $ \det[a \ b \ c] $	$2.842066937809649e-05 \text{ native}^3$
Node coefficient set	$\{-0.5, 0, +0.5\}$ on each basis vector
Integer node index set	$\{-1, 0, +1\}$ with coordinate $B(i,j,k)^T/2$
Node sheet index	$\lambda = i+j+k$

ICRS Cell3 base cell: 27 nodes and six boundary face planes

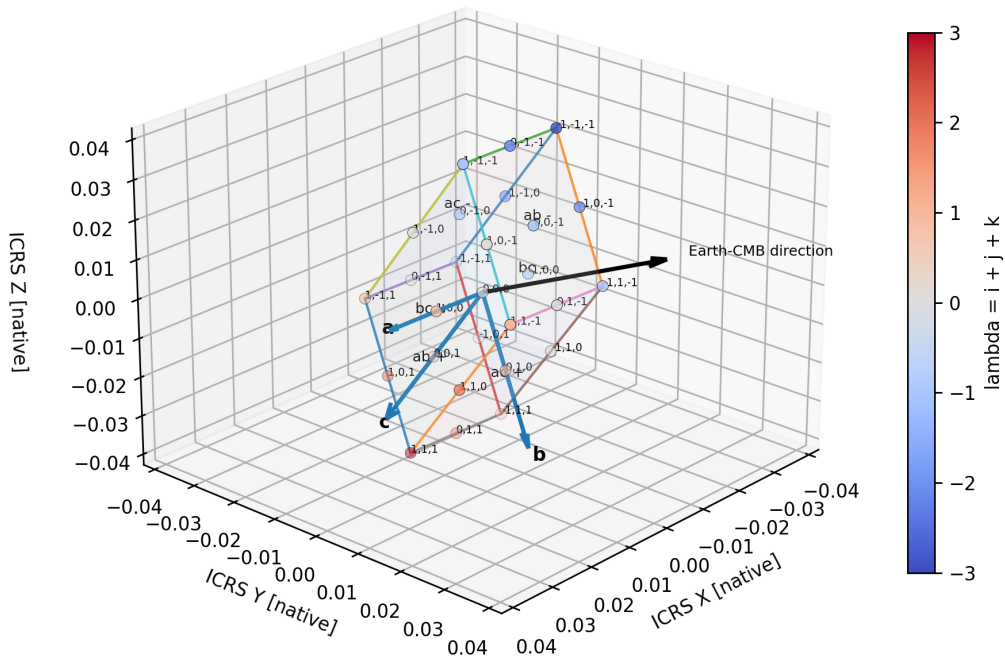


Figure 2. Dimensionally correct ICRS Cell3 base cell in native units. The twelve cell edges, all six boundary faces, all 27 nodes, each integer node label, the primitive basis vectors, and the Earth-CMB direction are generated from the source coordinates. Node color records $\lambda = i+j+k$.

3.1 Cell3 Face Areas, Normals, and Plane Equations

Each opposite face pair shares one unit normal. The plane convention used here is $n \cdot x + d = 0$. The two members of a pair use equal and opposite d . The labels ab , ac , and bc identify the two primitive basis vectors that lie within the corresponding face.

Face pair	Area [native ²]	Included angle [deg]	Unit normal ICRS	d values in $n \cdot x + d = 0$
A_ab	$7.547217896726240e-04$	83.933483012313	[0.1940922200808, 0.913177168821, 0.358379234791]	-1.882857350019305e-02, +1.882857350019305e-02
A_ac	$1.011255757688065e-03$	80.448130018338	[0.089052594102, 0.866576769291, -0.491033948326]	+1.405216690339142e-02, -1.405216690339142e-02

Face pair	Area [native^2]	Included angle [deg]	Unit normal ICRS	d values in $\mathbf{n} \cdot \mathbf{x} + d = 0$
A_bc	1.385784431733287e-03	50.363231813756	[-0.995965190309, 0.083602162507, -0.032619290564]	-1.025436161905391e-02, +1.025436161905391e-02

The Cell3 geometry does not enter the continuous-rim integral through a separate spatial scale. It enters through the ICRS orientation and the fixed phase metric G_{phase} supplied by the source model. The complete cell, node, vertex, and face-plane ledgers are retained because they define the authoritative orientation frame from which the Earth-CMB coupling is evaluated.

4. Flywheel Geometry and Continuous Mass Measure

For any axis orientation $\mathbf{n}$, the flywheel plane is the plane perpendicular to $\mathbf{n}$. Because $\mathbf{n}$ is constrained to be orthogonal to $\mathbf{c}_{\text{hat}}$, the Earth-CMB direction lies within the flywheel plane. A second in-plane unit vector is $\mathbf{h} = \mathbf{n} \times \mathbf{c}_{\text{hat}}$. The pair $\mathbf{c}_{\text{hat}}$ and $\mathbf{h}$ forms an orthonormal basis of the rim plane.

```

h(alpha) = n(alpha) cross c_hat
u(theta,alpha) = cos(theta) c_hat + sin(theta) h(alpha)
r(theta,alpha) = R u(theta,alpha)
t(theta,alpha) = n(alpha) cross u(theta,alpha)
dm = [M/(2 pi)] dtheta

```

The vector $\mathbf{u}$ is the outward radial unit vector from the flywheel center to the rim. The vector $\mathbf{t}$ is the positive tangent generated by the right-hand rule about $\mathbf{n}$. The local velocity is the sum of Earth-CMB translation and rim rotation. At the normalized reference $\omega = 1$ rad/s, the rim contribution has magnitude $R \omega = 0.5$ m/s, whereas the Earth-CMB translation magnitude is 369812.502 m/s.

Requested flywheel geometry at Earth center in the ICRS frame

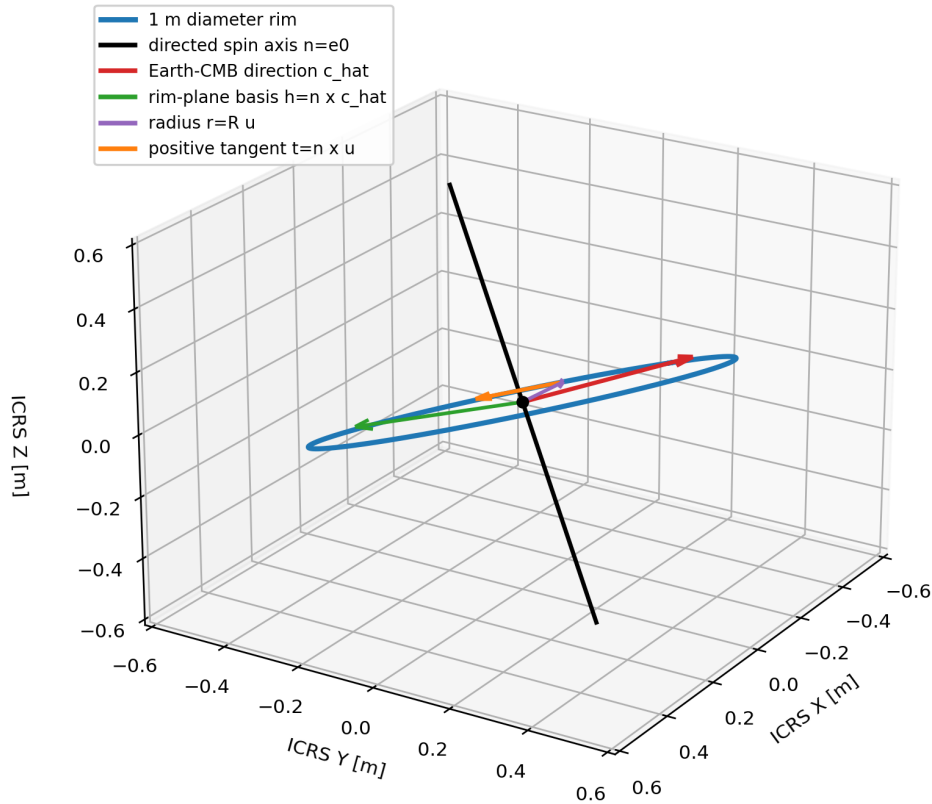


Figure 3. Exact one-metre-diameter rim at Earth center for $\alpha=0$, plotted in ICRS metres. The Earth-CMB direction lies in the rim plane, $n=e0$ is the directed spin axis, $h=n \times c_hat$ completes the rim-plane basis, and $t=n \times u$ is the local positive tangent. Direction arrows are scaled in length only for display; the rim radius is exactly 0.5 m.

5. Elemental K-Field Force Law

The source uses a fixed symmetric positive-definite phase metric G_phase . For a rim element, the translation coupling and rotational self-coupling are represented by two scalar kernels. The first is linear in the tangent and in the Earth-CMB translation state. The second is quadratic in the tangent.

$$\begin{aligned}
 g &= G_phase \ V_E \\
 B(\theta, \alpha) &= t(\theta, \alpha)^T g \\
 C(\theta, \alpha) &= t(\theta, \alpha)^T G_phase \ t(\theta, \alpha) \\
 dF &= -\kappa_F \ dm \ [\omega B + R \ \omega^2 C] \ u
 \end{aligned}$$

The bracketed quantity has dimensions of acceleration under the source normalization: B has dimensions of velocity, ωB has dimensions of acceleration, C is dimensionless, and $R \ \omega^2 C$ also has dimensions of acceleration.

Multiplication by dm gives force per unit κ_F . The source explicitly retains κ_F as an uncalibrated constitutive gain.

Consequently, every numerical force in this report is reported per unit kappa_F and scales linearly when kappa_F is assigned or measured.

G_phase row	X	Y	Z
1	2.976410637800	1.243150779500	-2.080360035400
2	1.243150779500	2.592228663500	-2.317883993700
3	-2.080360035400	-2.317883993700	6.773570913300

The three eigenvalues of G_phase are 1.429328753, 2.112545026, and 8.800336435. Their positivity verifies that the metric is positive-definite. Its anisotropy rotates and stretches the Earth-CMB velocity into g, producing the non-circular force envelope derived below.

6. Exact Continuous-Rim Integration

The full rim resultant is the integral of dF from theta=0 to 2 pi. The two terms behave differently under an antipodal shift by pi. At the antipodal point, u changes sign and t changes sign. Therefore B changes sign, while C does not.

```

u(theta+pi) = -u(theta)
t(theta+pi) = -t(theta)
B(theta+pi) = -B(theta)
C(theta+pi) = C(theta)

```

The rotational self-term is proportional to C u. Because C is even and u is odd under the antipodal shift, the two members of every antipodal pair cancel exactly. This removes the complete R omega-squared contribution from the net force, even though that term remains nonzero locally around the rim.

```

Integral_0^(2 pi) C(theta) u(theta) dtheta = 0

```

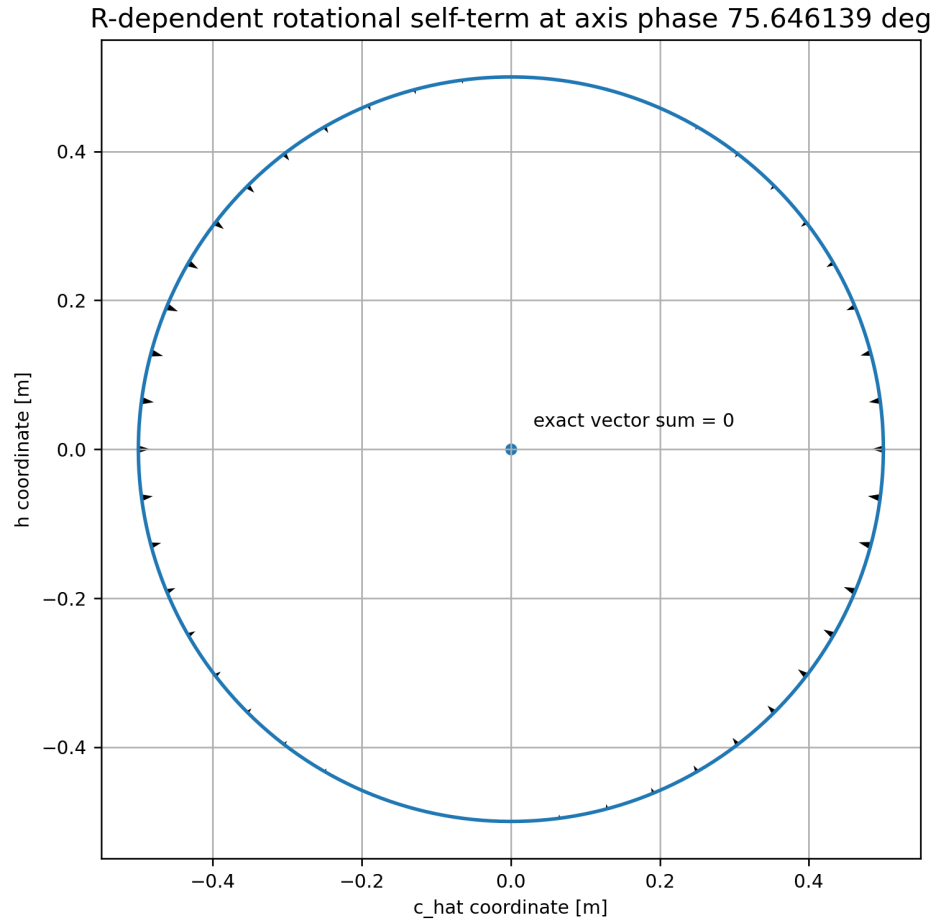


Figure 4. Actual R-dependent rotational self-term force density at a maximum-force axis orientation. Every local vector is radial. Opposite rim elements have equal scalar weight and opposite radial direction, so the complete vector integral is exactly zero. Arrow lengths are uniformly rescaled for visibility.

6.1 Surviving Translation-Coupled Integral

The translation term survives because both B and u change sign at the antipode; their product is even. The integral can be reduced without component-by-component trigonometric expansion. First use the scalar triple product to rewrite B .

$$B = (n \text{ cross } u) \text{ dot } g = u \text{ dot } (g \text{ cross } n)$$

$$\text{Integral } u B \, dm = [\text{Integral } u u^T \, dm] (g \text{ cross } n)$$

A uniform circular rim has the exact second directional moment $M/2$ in every direction within its plane and zero moment along the axis. Therefore the second-moment tensor is $M/2$ times the projector onto the rim plane.

$$\text{Integral } u u^T \, dm = (M/2)(I - n n^T)$$

$$(I - n n^T)(g \text{ cross } n) = g \text{ cross } n$$

Substitution into the translation term gives the exact continuous-rim resultant. The sign converts because minus $(g \text{ cross } n)$ equals $n \text{ cross } g$.

$$F(\alpha) = (\kappa_F M \omega / 2) n(\alpha) \text{ cross } [G_{\text{phase}} V_E]$$

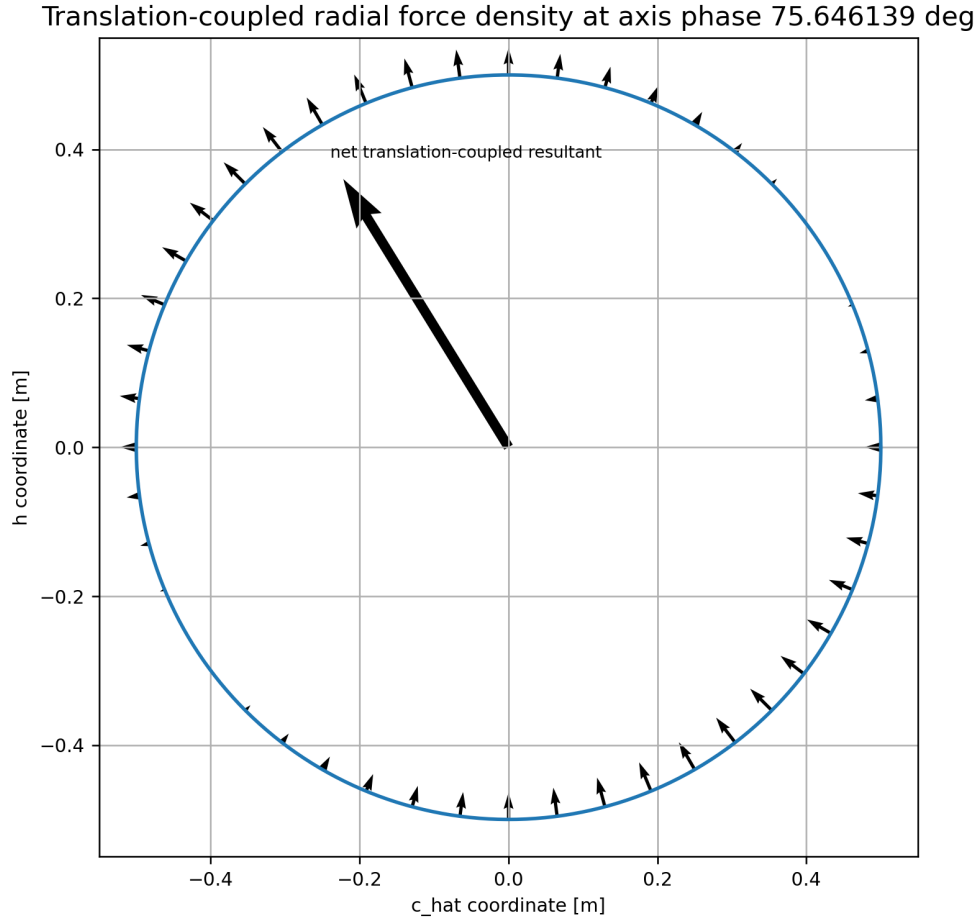


Figure 5. Actual translation-coupled radial force density at the axis orientation $\alpha=75.646139$ deg. The local vectors do not cancel pairwise. Their continuous vector sum is the net force shown from the origin. Arrow scales are stated only for visualization; all directions and relative magnitudes are computed from the source metric and velocity.

7. Closed-Form ICRS Force Equation

The axis is a first-harmonic rotation in the e_0 - A_0 plane, so the net force is also first harmonic in α . Evaluating the exact formula at $\alpha=0$ and $\alpha=90$ degrees gives the two ICRS coefficient vectors.

$$F(\alpha)/(\kappa_F \omega) = a_F \cos(\alpha) + b_F \sin(\alpha)$$

$$a_F = [6707.042413714, -361443.194211528, -173135.781944972] \text{ N}/(\kappa_F \text{ rad/s})$$

$$b_F = [150028.243956872, -217930.035903845, 378078.448087618] \text{ N}/(\kappa_F \text{ rad/s})$$

Thus every ICRS component is a sinusoid with the same fundamental sweep frequency but a different amplitude and phase. There are no higher harmonics in the exact continuous-rim resultant. Higher local harmonics present in the elemental force density disappear after the full rim integral.

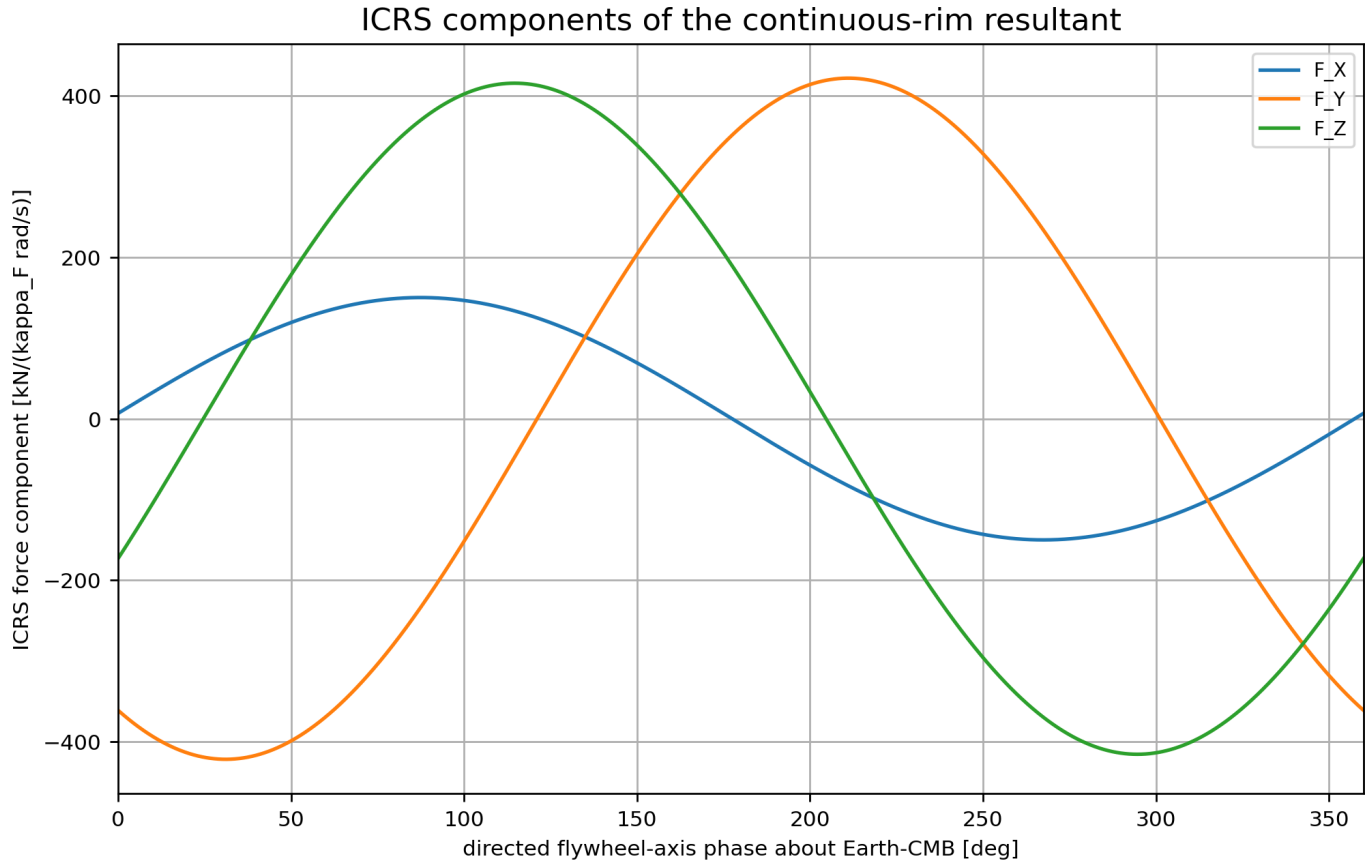


Figure 6. Exact ICRS force components over the complete directed axis sweep. Values are normalized by κ_F and ω . The closure at 360 degrees is exact.

8. Force Envelope as a Centered Ellipse

For every α , a cross product with $\mathbf{g}$ is perpendicular to $\mathbf{g}$. Therefore every force vector lies in the fixed plane through the origin whose normal is $\mathbf{g} = \mathbf{G}_{\text{phase}} \mathbf{V}_E$. The mapping $\mathbf{n} \rightarrow \mathbf{n} \times \mathbf{g}$ is linear. A unit circle in the axis plane is consequently mapped into an ellipse in the force plane.

```
F(alpha) dot [G_phase V_E] = 0
plane unit normal = [-0.945636157362, -0.154606293310, 0.286127859460]
plane-normal RA = 189.285391176 deg; Dec = 16.626277960 deg
```

The two singular values of the cross-product map restricted to the axis plane are $|\mathbf{g}|$ and $|\hat{\mathbf{c}} \cdot \mathbf{g}|$. After multiplication by $M/2$, these become the semi-major and semi-minor force magnitudes per $\kappa_F \omega$.

```
semi-major = (M/2) |g|
semi-minor = (M/2) |c_hat dot g|
semi-major = 465412.608635217417 N/(kappa_F rad/s)
semi-minor = 396230.434488575964 N/(kappa_F rad/s)
```

Continuous-rim net-force envelope in ICRS force space

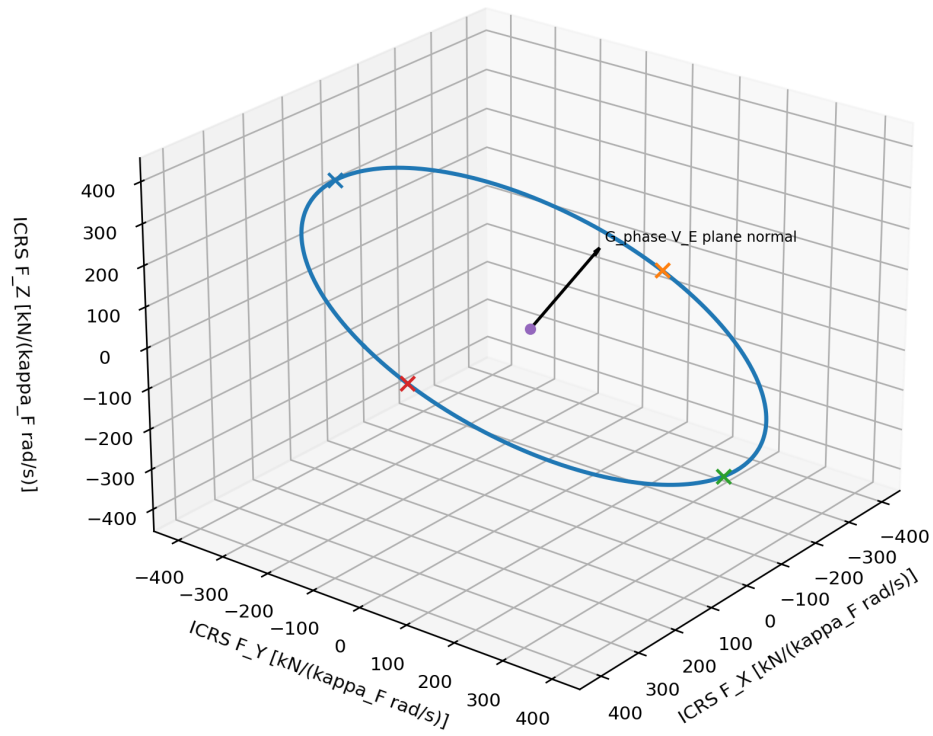


Figure 7. Full net-force vector envelope in ICRS force space. The closed locus is a centered ellipse. The black direction arrow is the plane normal $G_{\text{phase V}_E}$, scaled in force-space length only to make the orientation visible.

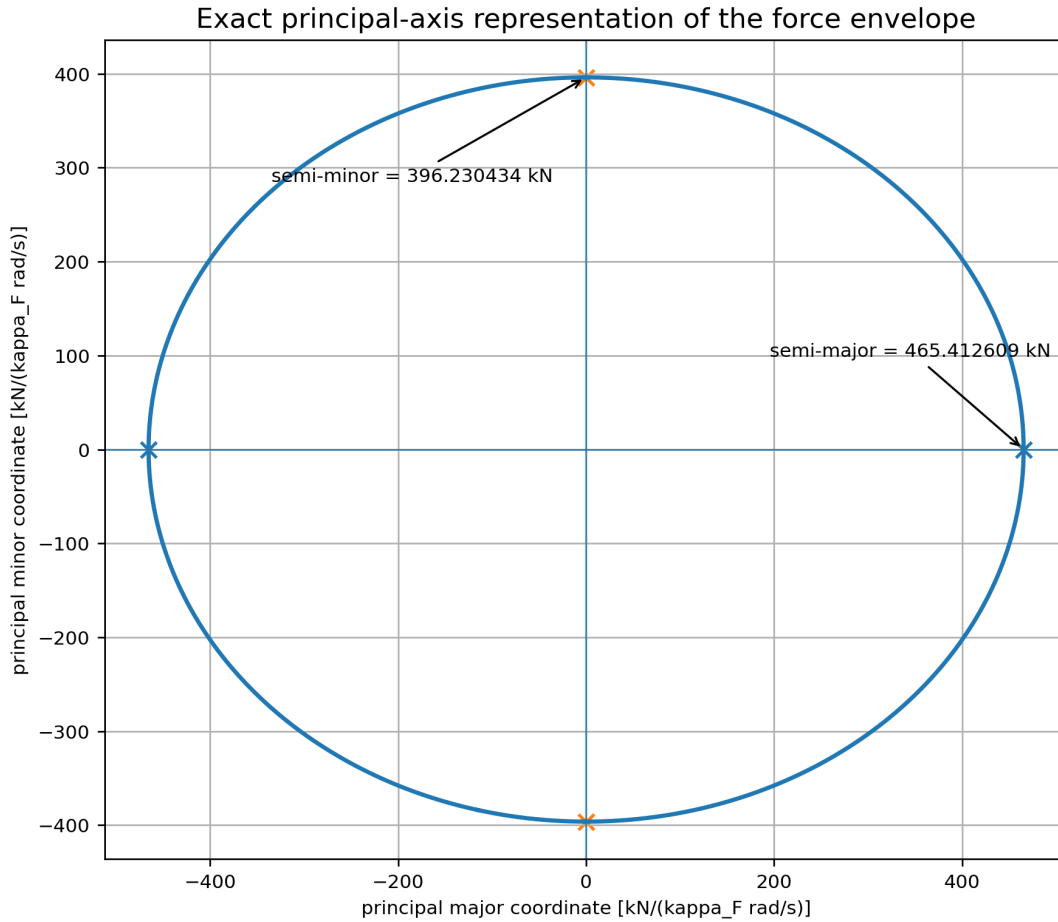


Figure 8. Dimensionally exact principal-axis view of the same ellipse. The axes are expressed in kN per (κ_F rad/s). The two labelled semi-axis values are computed analytically, not estimated from the sampled curve.

9. Earth-CMB-Resolved Force Decomposition

The orthonormal basis $\mathbf{e}_0$, $\mathbf{A}_0$, $\hat{\mathbf{c}}$ makes the physical structure transparent. Decompose $\mathbf{g}$ into the three scalar coefficients $p = \mathbf{e}_0 \cdot \mathbf{g}$, $q = \mathbf{A}_0 \cdot \mathbf{g}$, and $r = \hat{\mathbf{c}} \cdot \mathbf{g}$. The exact force then has two components perpendicular to the Earth-CMB direction and one component parallel to it.

Kernel coefficient	Definition	Value [m/s]
p	$\mathbf{e}_0 \cdot (\mathbf{G}_{\text{phase}} \mathbf{V}_E)$	473061.223213950288
q	$\mathbf{A}_0 \cdot (\mathbf{G}_{\text{phase}} \mathbf{V}_E)$	-121055.505194313228
r	$\hat{\mathbf{c}} \cdot (\mathbf{G}_{\text{phase}} \mathbf{V}_E)$	792460.868977151928
$\sqrt{p^2 + q^2}$	component of $\mathbf{g}$ perpendicular to $\hat{\mathbf{c}}$	488304.573239417106
$\mathbf{F}_{e0} = (\kappa_F M \omega / 2) r \sin(\alpha)$ $\mathbf{F}_{A0} = -(\kappa_F M \omega / 2) r \cos(\alpha)$ $\mathbf{F}_{CMB} = (\kappa_F M \omega / 2) [q \cos(\alpha) - p \sin(\alpha)]$		

The two transverse components form a circle of constant radius $|r|M/2$. Therefore the magnitude of the force perpendicular to the Earth-CMB direction is constant through the entire sweep. The parallel component is a single sinusoid with amplitude M

$\sqrt{p^2 + q^2}/2$. The changing parallel component converts the constant transverse circle into the total three-dimensional ellipse.

```
|F_perpendicular_CMB|/(kappa_F omega) = 396230.434488575964 N/(kappa_F rad/s)
max |F_parallel_CMB|/(kappa_F omega) = 244152.286619708553 N/(kappa_F rad/s)
RMS |F|/(kappa_F omega) = 432207.967008732783 N/(kappa_F rad/s)
```

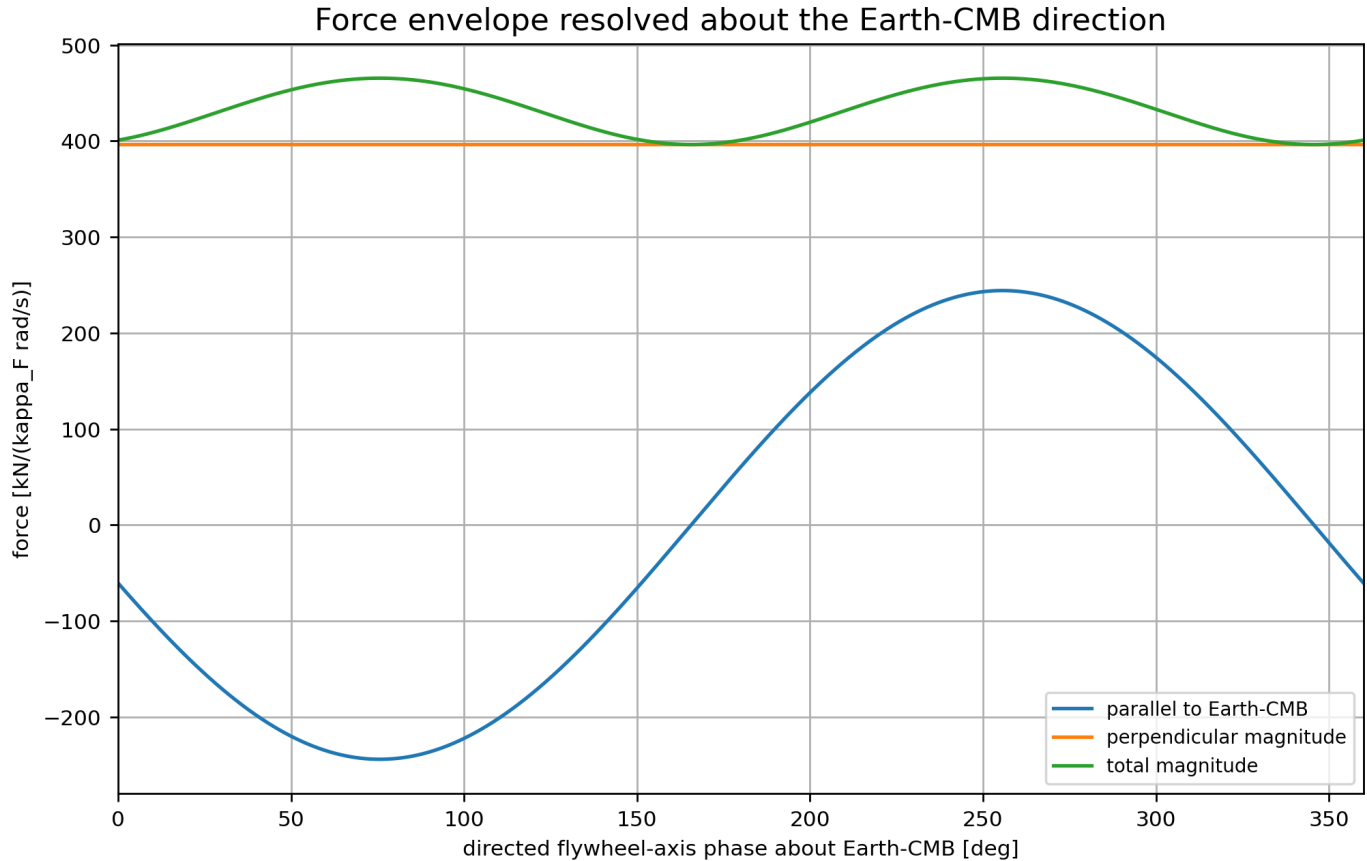


Figure 9. Total magnitude, Earth-CMB-parallel component, and perpendicular magnitude over the full sweep. The perpendicular magnitude is constant; the parallel component is sinusoidal and changes sign twice per 360-degree directed cycle.

10. Numerical Force Results for the Requested Flywheel

The following values use $M=1$ kg and the reference $\omega=1$ rad/s. They are normalized by κ_F , exactly as required by the source model. For any assigned spin rate and constitutive gain, multiply every entry by $\kappa_F \omega$, with ω expressed in rad/s.

Result	Value
Minimum total force	396230.434488575964 N/(kappa_F rad/s)
Maximum total force	465412.608635217417 N/(kappa_F rad/s)
Constant perpendicular-to-CMB magnitude	396230.434488575964 N/(kappa_F rad/s)
CMB-parallel amplitude	244152.286619708553 N/(kappa_F rad/s)
RMS total force	432207.967008732783 N/(kappa_F rad/s)
Full-sweep mean force vector	[0, 0, 0] exactly

Result	Value
Force-plane normal RA/Dec	189.285391176 deg, 16.626277960 deg
Cycle closure	F(360 deg)=F(0 deg)

At $\omega=1$ rad/s, the minimum and maximum normalized forces are 396.230434 kN and 465.412609 kN per unit κ_F . These large normalized values arise because the Earth-CMB translation state is hundreds of kilometres per second and is transformed by the anisotropic phase metric. Absolute engineering force requires the constitutive value of κ_F retained by the source model.

10.1 Linear Spin-Rate Scaling

Spin rate	ω [rad/s]	Minimum force / κ_F	Maximum force / κ_F	Parallel amplitude / κ_F
1 rpm	0.104719755120	41.493154 kN	48.737894 kN	25.567568 kN
10 rpm	1.047197551197	414.931541 kN	487.378944 kN	255.675677 kN
100 rpm	10.471975511966	4149.315407 kN	4873.789441 kN	2556.756767 kN
1000 rpm	104.719755119660	41493.154071 kN	48737.894406 kN	25567.567667 kN
Reference	1.000000000000	396.230434 kN	465.412609 kN	244.152287 kN

11. Extremal Axis Orientations and Force Directions

Maximum force occurs when the flywheel axis is perpendicular to $\mathbf{g}$. Minimum force occurs when the axis is parallel or antiparallel to the projection of $\mathbf{g}$ into the plane orthogonal to $\mathbf{c}_{\text{hat}}$. The four directed orientations below provide the complete pair of maximum points and the complete pair of minimum points.

Condition	Axis phase [deg]	Axis RA/Dec [deg]	Force RA/Dec [deg]	F	F_parallel_CMB
Maximum force; negative CMB-parallel component	75.646138730634	84.082114314, 41.289922334	296.050867377, 44.008760899	465412.608635217	-244152.286619709
Minimum force; zero CMB-parallel component	165.646138730634	70.207297827, -47.867366475	84.082114314, 41.289922334	396230.434488576	0.000000000
Maximum force; positive CMB-parallel component	255.646138730634	264.082114314, -41.289922334	116.050867377, -44.008760899	465412.608635217	244152.286619709
Minimum force; zero CMB-parallel component	345.646138730634	250.207297827, 47.867366475	264.082114314, -41.289922334	396230.434488576	0.000000000

The maximum-force phases are 75.646138731 degrees and 255.646138731 degrees. The minimum-force phases are 165.646138731 degrees and 345.646138731 degrees. The two members of each pair are antipodal in axis orientation and produce opposite force vectors of equal magnitude.

12. Full-Sweep Vector Locus and Directional Great Circle

Although the force magnitude varies elliptically, every nonzero force vector lies in one plane through the origin. Normalizing the vector removes the varying magnitude and places the force direction on the unit sphere. The resulting direction locus is a great circle whose pole is $\mathbf{g}_{\text{hat}}$. The directional sweep therefore has an exact spherical representation in addition to its force-space ellipse.

ICRS direction locus of the net force: a great circle

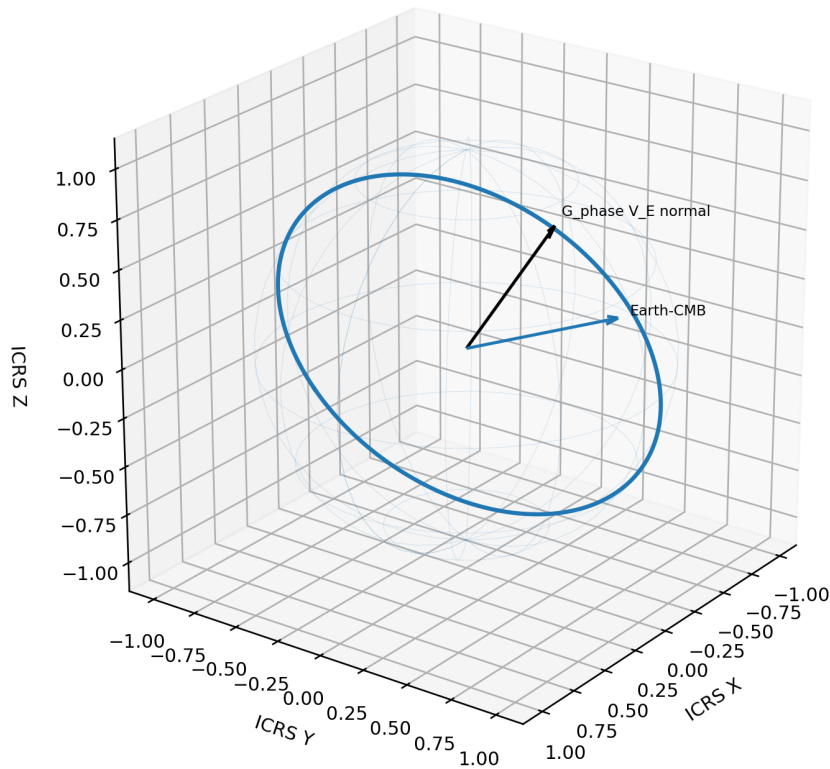


Figure 10. Unit-sphere ICRS direction locus of the continuous-rim resultant. Because all force vectors satisfy $F \cdot (G_{\text{phase}} V_E) = 0$, their normalized directions trace the great circle perpendicular to $G_{\text{phase}} V_E$. The Earth-CMB direction is shown for reference.

The force never vanishes for this source metric and translation state because the semi-minor value is nonzero. A zero-force orientation would require an allowed axis n parallel to g . That can occur only when g lies entirely in the axis-sweep plane, equivalent to $c_{\text{hat}} \cdot g = 0$. Here $c_{\text{hat}} \cdot g$ is 792460.868977 m/s, so the ellipse does not pass through the origin.

13. Radius, Mass, and Spin Scaling

The requested radius appears in the local velocity and in the local rotational self-term, but it disappears from the integrated resultant at fixed ω . This does not mean radius is irrelevant to every operating comparison. It means that the exact force law depends on $M \omega$ after the uniform-rim integration.

Constraint held fixed	Resulting force scaling
M, ω , and κ_F	F is independent of rim radius R
R, ω , and κ_F	F is linear in total mass M
M, R, and κ_F	F is linear in angular speed ω
M, rim speed $v_{\text{rim}} = R \omega$	F is proportional to $1/R$
Angular momentum $L = M R^2 \omega$	$F = (\kappa_F L / 2R^2) n \times g$
Rotational energy $E_{\text{rot}} = 0.5 M R^2 \omega^2$	F magnitude scale is proportional to $\sqrt{M E_{\text{rot}}} / R$
Spin reversal $\omega \rightarrow -\omega$	Every force vector reverses; the ellipse is unchanged as a set

For the specified one-kilogram thin rim, the moment of inertia is 0.25 kg m squared. At the reference spin rate the rim stores 0.125 J and 0.25 kg m squared per second of angular momentum. These rotational quantities change with radius even though the fixed-omega net force does not.

14. Torque, Internal Stress, and Resultant Interpretation

Every elemental force is parallel or antiparallel to the local radial vector u . Its moment arm is $r=R u$. Therefore the elemental torque about the flywheel center vanishes before integration.

```
d tau = r cross dF
r = R u and dF = scalar times u
d tau = 0 for every theta
tau_net = 0
```

The K-Field resultant is consequently a pure net force at the center-of-mass level under this elemental law, not a driving or braking torque on the rim. Local radial loads remain and would be carried as hoop stress and bearing reaction in a physical structure. The rotational self-term contributes to those local loads while making no contribution to the net resultant.

A complete directed sweep has zero mean force because $n(\alpha)$ averages to zero and the cross-product map is linear. A static orientation can have a nonzero normalized force, while an equal-time uniform sweep over all orientations has no net cycle-average force. Nonuniform dwell or controlled orientation scheduling would weight the ellipse unevenly; such control laws are outside the present fixed-sweep calculation but can be constructed directly from the closed-form vector equation.

15. Numerical Reconstruction and Validation

The closed form was reconstructed independently by numerical integration of 16384 uniformly spaced rim elements at 17 independent axis orientations. The maximum vector difference between the analytic and numerical results is 3.494884e-09 N per (κ_F rad/s). This residual is at floating-point summation scale relative to a force magnitude of approximately 4.0e5 N per (κ_F rad/s).

Validation quantity	Result
G_phase symmetry maximum absolute residual	0.000000e+00
G_phase eigenvalues	1.429328753487, 2.112545026447, 8.800336434665
Sweep-frame Gram residual	2.220446e-16
Sweep-frame determinant	1.000000000000000
Earth-CMB speed reconstruction residual	0.000000e+00 m/s
Cell basis-length maximum residual	0.000000e+00 native
Cell volume residual	0.000000e+00 native^3
27-node reconstruction maximum residual	0.000000e+00 native
Analytic versus 16384-element quadrature	3.494884e-09 N/(κ_F rad/s)
Force-plane residual max F dot g	3.644335e-05
360-degree closure residual	1.372825e-10 N/(κ_F rad/s)
Numerical sweep mean-force magnitude	1.939450e-10 N/(κ_F rad/s)

The sampled minimum and maximum agree with the analytic semi-axis values. The force-plane residual verifies that the full sampled ellipse remains perpendicular to g . The node reconstruction verifies that every listed Cell3 node equals $B(i,j,k)/2$ to binary64 precision.

16. Worked Reconstruction Example

Consider $\alpha=0$. The directed axis is $n=e_0$. The exact force is one-half of e_0 cross g for $M=1$ and $\omega=1$. This is the cosine coefficient of the complete ICRS harmonic equation.

$$\begin{aligned} n(0) &= e_0 = [-0.200864884110, -0.426231479950, 0.882031758970] \\ g &= [-880221.981635091, -143911.436561599, 266335.026948619] \text{ m/s} \\ F(0)/(\kappa_F \omega) &= [6707.042413714, -361443.194211528, -173135.781944972] \text{ N}/(\kappa_F \text{ rad/s}) \\ |F(0)|/(\kappa_F \omega) &= 400826.852954575 \text{ N}/(\kappa_F \text{ rad/s}) \\ F_{\text{parallel_CMB}}(0)/(\kappa_F \omega) &= -60527.752597157 \text{ N}/(\kappa_F \text{ rad/s}) \end{aligned}$$

At $\alpha=75.646138731$ degrees, the axis is perpendicular to g and the total force reaches the semi-major value. The corresponding force has a negative Earth-CMB-parallel component of magnitude 244152.286620 N per $(\kappa_F \text{ rad/s})$. Advancing the directed axis by 180 degrees reverses both the axis and the force while preserving the force magnitude.

17. Principal Findings

Finding	Exact statement
Continuous-rim closure	The full zero-thickness rim integrates analytically; no point-mass approximation is required.
Self-term cancellation	The complete $R \omega^2 C u$ contribution cancels by antipodal odd symmetry.
Surviving resultant	$F=(\kappa_F M \omega/2)n \text{ cross}(G_{\text{phase}} V_E)$.
Envelope geometry	A centered ellipse in the plane perpendicular to $G_{\text{phase}} V_E$.
CMB transverse component	Constant magnitude 396230.434489 N/ $(\kappa_F \text{ rad/s})$.
CMB axial component	Sinusoidal with amplitude 244152.286620 N/ $(\kappa_F \text{ rad/s})$.
Total force range	396230.434489 to 465412.608635 N/ $(\kappa_F \text{ rad/s})$.
Torque	Zero locally and globally about the flywheel center.
Radius at fixed ω	No dependence for a uniform zero-thickness rim.
Full-cycle mean	Exactly zero for uniform dwell in directed axis phase.
Spin reversal	Reverses every vector while preserving the ellipse.
Directional locus	The normalized force direction traces a great circle with pole $G_{\text{phase}} V_E$.

The central structural result is that continuous mass distribution does not erase the K-Field translation coupling. It removes the radius-dependent rotational self-resultant and leaves a simpler exact vector law. The force envelope is controlled entirely by the total mass, spin rate, constitutive gain, directed axis orientation, and the metric-transformed Earth-CMB velocity.

18. Conclusion

A one-kilogram continuous rim with a one-metre diameter, translating with the Earth-CMB state and spinning about an axis orthogonal to that state, has an exact K-Field resultant under the attached source law. The continuous integration converts the elemental radial-force field into the closed form $F=(\kappa_F M \omega/2)n \text{ cross}(G_{\text{phase}} V_E)$. The full directed axis sweep maps the circular set of allowed axes into a centered force ellipse. For the authoritative ICRS metric and Earth-CMB vector, the normalized ellipse has semi-major 465412.608635 N and semi-minor 396230.434489 N per $(\kappa_F \text{ rad/s})$. The transverse-to-CMB magnitude is constant, the CMB-parallel component is sinusoidal, the mean over a uniform full sweep is zero, and the torque about the flywheel center is zero. These relations provide a complete analytic reference for further orientation scheduling, material stress analysis, multi-flywheel superposition, and constitutive calibration.

The companion JSON preserves the exact source constants, Cell3 geometry, all nodes and planes, full 0.1-degree force ledger, extrema, formulas, scaling laws, and validation quantities required to reproduce every numerical statement and figure in this report.

Appendix A. Complete 27-Node Cell3 Ledger

Every node is indexed by (i,j,k) in $\{-1,0,+1\}$ cubed. The coefficient vector is one-half of that integer index, the ICRS coordinate is $B(i,j,k)^{AT/2}$, and $\lambda=i+j+k$.

No.	i	j	k	lambda	X native	Y native	Z native
1	-1	-1	-1	-3	-0.010974902499961	0.007385983127423	0.039661911011939
2	-1	-1	0	-2	-0.011290092242635	-0.004742877881141	0.018199733726392
3	-1	-1	1	-1	-0.011605281985309	-0.016871738889705	-0.003262443559155
4	-1	0	-1	-2	-0.009875529515800	0.013837068256769	0.022628660243911
5	-1	0	0	-1	-0.010190719258473	0.001708207248205	0.001166482958364
6	-1	0	1	0	-0.010505909001147	-0.010420653760359	-0.020295694327182
7	-1	1	-1	-1	-0.008776156531638	0.020288153386115	0.005595409475884
8	-1	1	0	0	-0.009091346274312	0.008159292377552	-0.015866767809663
9	-1	1	1	1	-0.009406536016985	-0.003969568631012	-0.037328945095210
10	0	-1	-1	-2	-0.000784183241488	0.005677775879218	0.038495428053575
11	0	-1	0	-1	-0.001099372984162	-0.006451085129346	0.017033250768028
12	0	-1	1	0	-0.001414562726835	-0.018579946137910	-0.004428926517519
13	0	0	-1	-1	0.000315189742674	0.012128861008564	0.021462177285547
14	0	0	0	0	0.000000000000000	0.000000000000000	0.000000000000000
15	0	0	1	1	-0.000315189742674	-0.012128861008564	-0.021462177285547
16	0	1	-1	0	0.001414562726835	0.018579946137910	0.004428926517519
17	0	1	0	1	0.001099372984162	0.006451085129346	-0.017033250768028
18	0	1	1	2	0.000784183241488	-0.005677775879218	-0.038495428053575
19	1	-1	-1	-1	0.009406536016985	0.003969568631012	0.037328945095210
20	1	-1	0	0	0.009091346274312	-0.008159292377552	0.015866767809663
21	1	-1	1	1	0.008776156531638	-0.020288153386115	-0.005595409475884
22	1	0	-1	0	0.010505909001147	0.010420653760359	0.020295694327182
23	1	0	0	1	0.010190719258473	-0.001708207248205	-0.001166482958364
24	1	0	1	2	0.009875529515800	-0.013837068256769	-0.022628660243911
25	1	1	-1	1	0.011605281985309	0.016871738889705	0.003262443559155
26	1	1	0	2	0.011290092242635	0.004742877881141	-0.018199733726392
27	1	1	1	3	0.010974902499961	-0.007385983127423	-0.039661911011939

Appendix B. Cell3 Vertices and Boundary Plane Ledger

Vertex	a coefficient	b coefficient	c coefficient	X native	Y native	Z native
1	-0.5	-0.5	-0.5	-0.010974902499961	0.007385983127423	0.039661911011939
2	-0.5	-0.5	+0.5	-0.011605281985309	-0.016871738889705	-0.003262443559155
3	-0.5	+0.5	-0.5	-0.008776156531638	0.020288153386115	0.005595409475884
4	-0.5	+0.5	+0.5	-0.009406536016985	-0.003969568631012	-0.037328945095210
5	+0.5	-0.5	-0.5	0.009406536016985	0.003969568631012	0.037328945095210
6	+0.5	-0.5	+0.5	0.008776156531638	-0.020288153386115	-0.005595409475884
7	+0.5	+0.5	-0.5	0.011605281985309	0.016871738889705	0.003262443559155
8	+0.5	+0.5	+0.5	0.010974902499961	-0.007385983127423	-0.039661911011939

Plane	Unit normal ICRS	d	Equation
ab faces (constant c coefficient) #1	[0.194092200808, 0.913177168821, 0.358379234791]	-1.882857350019305e-02	$(+0.194092200808)X + (+0.913177168821)Y + (+0.358379234791)Z + (-1.882857350019305e-02) = 0$
ab faces (constant c coefficient) #2	[0.194092200808, 0.913177168821, 0.358379234791]	+1.882857350019305e-02	$(+0.194092200808)X + (+0.913177168821)Y + (+0.358379234791)Z + (+1.882857350019305e-02) = 0$
ac faces (constant b coefficient) #1	[0.089052594102, 0.866576769291, -0.491033948326]	+1.405216690339142e-02	$(+0.089052594102)X + (+0.866576769291)Y + (-0.491033948326)Z + (+1.405216690339142e-02) = 0$
ac faces (constant b coefficient) #2	[0.089052594102, 0.866576769291, -0.491033948326]	-1.405216690339142e-02	$(+0.089052594102)X + (+0.866576769291)Y + (-0.491033948326)Z + (-1.405216690339142e-02) = 0$
bc faces (constant a coefficient) #1	[-0.995965190309, 0.083602162507, -0.032619290564]	-1.025436161905391e-02	$(-0.995965190309)X + (+0.083602162507)Y + (-0.032619290564)Z + (-1.025436161905391e-02) = 0$
bc faces (constant a coefficient) #2	[-0.995965190309, 0.083602162507, -0.032619290564]	+1.025436161905391e-02	$(-0.995965190309)X + (+0.083602162507)Y + (-0.032619290564)Z + (+1.025436161905391e-02) = 0$

Appendix C. Formula and Identity Ledger

Identifier	Formula or identity
C01	$\mathbf{n}(\alpha) = \cos(\alpha)\mathbf{e}_0 + \sin(\alpha)\mathbf{A}_0$
C02	$\mathbf{h}(\alpha) = \mathbf{n}(\alpha) \times \mathbf{c}_{\text{hat}}$
C03	$\mathbf{u}(\theta, \alpha) = \cos(\theta)\mathbf{c}_{\text{hat}} + \sin(\theta)\mathbf{h}(\alpha)$
C04	$\mathbf{t}(\theta, \alpha) = \mathbf{n}(\alpha) \times \mathbf{u}(\theta, \alpha)$
C05	$\mathbf{r}(\theta, \alpha) = R \mathbf{u}(\theta, \alpha)$
C06	$\mathbf{v}(\theta, \alpha) = \mathbf{V}_E + R \omega \mathbf{t}(\theta, \alpha)$
C07	$\mathbf{B} = \mathbf{t}^T \mathbf{G}_{\text{phase}} \mathbf{V}_E$
C08	$\mathbf{C} = \mathbf{t}^T \mathbf{G}_{\text{phase}} \mathbf{t}$
C09	$d\mathbf{m} = (M/2\pi)d\theta$
C10	$d\mathbf{F} = -\kappa_F d\mathbf{m} [\omega \mathbf{B} + R \omega^2 \mathbf{C}] \mathbf{u}$
C11	$\mathbf{u}(\theta + \pi) = -\mathbf{u}(\theta); \mathbf{t}(\theta + \pi) = -\mathbf{t}(\theta)$
C12	$\mathbf{B}(\theta + \pi) = -\mathbf{B}(\theta); \mathbf{C}(\theta + \pi) = \mathbf{C}(\theta)$
C13	$\int \mathbf{C} \mathbf{u} d\theta = 0$
C14	$\int \mathbf{u}^T d\mathbf{m} = (M/2)(1 - n^T)$
C15	$\mathbf{F} = (\kappa_F M \omega/2) n \times (\mathbf{G}_{\text{phase}} \mathbf{V}_E)$
C16	$\boldsymbol{\tau} = \int \mathbf{r} \times d\mathbf{F} = 0$
C17	$\mathbf{F} \cdot (\mathbf{G}_{\text{phase}} \mathbf{V}_E) = 0$
C18	$\text{semi-major} = (\kappa_F M \omega /2) \mathbf{G}_{\text{phase}} \mathbf{V}_E $
C19	$\text{semi-minor} = (\kappa_F M \omega /2) \mathbf{c}_{\text{hat}} \cdot \mathbf{G}_{\text{phase}} \mathbf{V}_E $
C20	$ \mathbf{F}_{\text{perpendicular_CMB}} = \text{semi-minor}$
C21	$\max \mathbf{F}_{\text{parallel_CMB}} = (\kappa_F M \omega /2) \mathbf{c}_{\text{hat}} \times \mathbf{G}_{\text{phase}} \mathbf{V}_E $
C22	$\langle \mathbf{F} \rangle \text{ over uniform full } \alpha \text{ sweep} = 0$

Appendix D. Fifteen-Degree Force Sweep Ledger

The machine-readable companion contains the complete 0.1-degree ledger. This appendix provides a compact fifteen-degree reference table. All forces are in N per (kappa_F rad/s).

alpha	F_X	F_Y	F_Z	F	F_parallel_CMB	F_perp_CMB
0.0	6707.042414	-361443.194212	-173135.781945	400826.852955	-60527.752597	396230.434489
15.0	45308.672325	-405531.759817	-69382.420328	413911.589906	-119683.946474	396230.434489
30.0	80822.591093	-421984.006164	39099.238575	431429.650287	-170683.877187	396230.434489
45.0	110828.583844	-409678.739853	144916.348984	448464.483640	-210051.983737	396230.434489
60.0	133281.791759	-369454.544446	240857.649695	460731.053699	-235105.394723	396230.434489
75.0	146652.065824	-304052.632388	320384.899615	465404.464497	-244136.761589	396230.434489
90.0	150028.243957	-217930.035904	378078.448088	461459.952154	-236530.611607	396230.434489
105.0	143180.245198	-116955.867619	410006.575127	449760.657720	-212805.291330	396230.434489
120.0	126574.749345	-8011.350235	413993.431640	432984.884661	-174577.642126	396230.434489
135.0	101343.393499	101479.127429	389767.319943	415315.615412	-124452.815114	396230.434489
150.0	69205.652864	204053.970260	338979.209512	401664.473969	-65846.734420	396230.434489
165.0	32351.661354	292722.872234	265090.226143	396240.000401	-2753.307593	396230.434489
180.0	-6707.042414	361443.194212	173135.781945	400826.852955	60527.752597	396230.434489
195.0	-45308.672325	405531.759817	69382.420328	413911.589906	119683.946474	396230.434489
210.0	-80822.591093	421984.006164	-39099.238575	431429.650287	170683.877187	396230.434489
225.0	-110828.583844	409678.739853	-144916.348984	448464.483640	210051.983737	396230.434489
240.0	-133281.791759	369454.544446	-240857.649695	460731.053699	235105.394723	396230.434489
255.0	-146652.065824	304052.632388	-320384.899615	465404.464497	244136.761589	396230.434489
270.0	-150028.243957	217930.035904	-378078.448088	461459.952154	236530.611607	396230.434489
285.0	-143180.245198	116955.867619	-410006.575127	449760.657720	212805.291330	396230.434489
300.0	-126574.749345	8011.350235	-413993.431640	432984.884661	174577.642126	396230.434489
315.0	-101343.393499	-101479.127429	-389767.319943	415315.615412	124452.815114	396230.434489
330.0	-69205.652864	-204053.970260	-338979.209512	401664.473969	65846.734420	396230.434489
345.0	-32351.661354	-292722.872234	-265090.226143	396240.000401	2753.307593	396230.434489
360.0	6707.042414	-361443.194212	-173135.781945	400826.852955	-60527.752597	396230.434489

Appendix E. Source Lock and Reconstruction Metadata

Field	Value
Source file	K-Field_dual_axis_kinetic_mass_CMB_traversal_lateral_force_comptation_1784803951(1).json
Source schema	K-Field.dual_axis_CMB_traversal_lateral_force_computation.v1
Source epoch	1784803951
Report epoch	1784808116
Coordinate frame	ICRS
Metric lock	fixed symmetric positive-definite G_phase
Preferred translation state	Earth-CMB vector V_E
Constitutive normalization	kappa_F is explicit and uncalibrated in absolute engineering units
Output interpretation	all force magnitudes are per declared native normalization and scale linearly with kappa_F
Machine-readable companion	K-Field_flywheel_force_analysis_1784808116.json
Sweep resolution in companion	0.1 degree including 0 and 360 degrees
Figure generation	All figures generated from the numerical arrays stored in the source and companion JSON files
PDF generator	Python ReportLab using only Helvetica, Helvetica-Bold, Times-Roman, and Courier core fonts

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